

CSCE 628 COMPUTATIONAL BIOLOGY · MODULE 05 · LECTURE NOTES

Mapping and Assembly

Reconstructing from fragments, and where repeats break everything.

MODULE OUTCOMES

- Explain read mapping and its error modes.
- Explain de Bruijn graph assembly.
- Explain why repeats are the fundamental obstacle.
- Explain coverage and why it must be uneven.
- Interpret assembly quality metrics critically.

1 Mapping

MAPPING PLACES EACH READ ON A REFERENCE, AND THE HARD CASES ARE THE ONES THAT MATTER

Hundreds of millions of short reads against a single fixed reference — which is Module 04's many-queries case in its purest form, and is exactly what the FM-index was adopted for (Module 04 §2).

And most reads map uniquely and easily — the problem is entirely the minority that do not: reads from repetitive regions, reads from sequence absent in the reference, and reads spanning structural variants, all of which are the biologically interesting cases.

A multi-mapping read is usually either assigned arbitrarily among its best positions or discarded — and both choices bias the result against repetitive regions systematically, so coverage appears low exactly where the mapping is hard.

Plus reference bias: a read carrying a genuine variant aligns slightly worse than one matching the reference, so it is more likely to fall below a mapping quality threshold — and variants are therefore systematically under-detected, which is a measured effect and a serious one, especially for the alleles least represented in the reference (§4's closing callout).

2 Assembly

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THE GRAPH
  nodes: every (k-1)-mer seen in the reads
  edges: every k-mer, joining its prefix to its
         suffix
  an assembly is a path using each edge once --
  an Eulerian path, which is tractable
  (a Hamiltonian path on the overlap graph, the
  older formulation, is not)

WHY k MATTERS
  small k   more overlaps found, and more false
            ones; the graph is tangled
  large k   cleaner graph, but needs higher
            coverage and loses low-coverage
            regions entirely

WHAT BREAKS IT
  sequencing errors create spurious nodes --
            removed by coverage thresholds
  repeats longer than k create ambiguous paths
            that NO amount of data resolves (section 3)
  uneven coverage makes the error threshold a
            guess rather than a calculation

SO THE OUTPUT IS CONTIGS, not a genome: pieces that
the data could order in many different ways.
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The output is contigs, not a genome — pieces that the data could order in many ways, and the ordering is a separate inference requiring other evidence (paired reads, optical maps, linkage, synteny with a relative). The reformulation from Hamiltonian paths on an overlap graph to Eulerian paths on a de Bruijn graph is one of the genuinely elegant moves in the field: the same problem, restated so that a polynomial algorithm applies (CSCE 629 Module 11's reductions, used constructively).

3 Repeats

A REPEAT LONGER THAN YOUR READ LENGTH IS NOT RESOLVABLE BY MORE READS OF THAT LENGTH

If the same sequence appears twice in the genome and is longer than a single read, then no read spans from unique sequence through the repeat into unique sequence on the other side — so the connection simply is not present in the data, and no algorithm can recover it.

Which is an information-theoretic limit rather than an algorithmic one — more coverage does not help at all, since every additional read has the same property — and this is the clearest such limit in the course, worth recognising as a type.

And roughly half of a human genome is repetitive to some degree, which is why short-read assembly of mammalian genomes remained poor for a decade despite enormous improvements in cost and throughput: the throughput was not the binding constraint.

So the fix is longer reads, or paired reads with a known separation that spans the repeat — changing the measurement rather than the algorithm — which is this semester's recurring shape: when the model comes from outside, the lever is frequently outside too (CSCE 640 Module 13 §3's model-from-outside framing).

4 Reading the metrics

- **N50: the contig length such that half the total assembly length lies in contigs at least that long — the standard reported metric, and it rewards aggressive joining** whether or not the joins are correct.
- **So a wrong assembly can have a better N50 than a correct one — which makes it a proxy with the usual proxy problem** (CSCE 676 Module 13, CSCE 671 Module 12 §4): the metric is optimised, the thing it stood for is not.
- **Total assembly length compared against the expected genome size — substantially too large means uncollapsed duplicate haplotypes, too small means collapsed repeats — and both are diagnosable.**
- **And completeness measured by recovery of conserved single-copy genes — which measures something you actually care about, namely whether the genes are there — and is the metric to prefer** when reporting.
- **Plus the independent check: does the assembly agree with a measurement that did not go through it?** — optical maps, genetic linkage, or synteny with a related species. **A wrong assembly can have a better N50 than a correct one, because N50 rewards joining — which is the proxy problem in its sharpest form** and a good example to keep.

AND THE HONEST SUMMARY

A reference genome is an assembly, with everything in this module behind it — **which is Module 01 §1's 'construct'**, now with the construction visible and the sources of error nameable.

And reference bias means your variant calls are biased toward the reference (§1) — **which matters most for the populations least represented in the reference's source individuals, and that is a known, measured, and consequential problem** in medical genomics rather than a theoretical concern.

Which is why pangenome references exist: a graph representing variation rather than a single linear sequence — **which is the current direction of the field and is substantially harder computationally**, since every algorithm in Modules 02 to 05 assumed a string.

So the thing to carry forward: your coordinates refer to a particular assembly of a particular set of individuals — **name it, which is Module 12 §2's requirement** and the single cheapest thing you can do for reproducibility.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Compeau, Pevzner & Tesler — How to apply de Bruijn graphs to genome assembly (free)	§2 — four pages, and the clearest explanation of the reformulation.
Li & Durbin — Fast and accurate short read alignment with Burrows-Wheeler transform (free)	§1 — the mapper most widely used, and its handling of the hard cases.
Nagarajan & Pop — Sequence assembly demystified (free)	§§2 to 4 — including an honest treatment of the metrics.
The Human Pangenome Reference Consortium (free)	§4's closing callout — what replaces a single linear reference, and why.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Map a small read set** to a small reference and count the multi-mappers.
- 2 **Inject a variant** into simulated reads and measure the mapping rate difference.
- 3 **Build a de Bruijn graph by hand** for a short string at two values of k .

- 4 **Implement the graph** and find an Eulerian path.
- 5 **Add a simulated repeat** longer than k and observe the ambiguity.
- 6 **Add more coverage** and confirm it does not help.
- 7 **Add errors** and find a coverage threshold that removes them.
- 8 **Compute N50** for two assemblies, one deliberately over-joined.
- 9 **Compute completeness** by a set of expected genes.
- 10 **Find which reference build** a public variant file uses.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What makes mapping easy, and which cases are hard?
- 2 How are multi-mappers handled, and what does that bias?
- 3 State reference bias and why it matters.
- 4 Describe the de Bruijn construction and why it is tractable.
- 5 What does k trade off?
- 6 Name three things that break assembly.
- 7 What is the output, actually?
- 8 Why is a long repeat an information-theoretic limit?
- 9 What is the fix, and what kind of fix is it?
- 10 Why is N50 a poor metric, and what is better?